Treatment following ROSC

- Ensure an adequate airway and adequate breathing:
  - Ventilate to an $ETCO_2$ of 35-45 mmHg if an ETT or SGA has been placed.
  - If the patient is ventilated via an ETT, administer sedation and neuromuscular blockade if required, using the **'post intubation'** guideline.
- Ensure an adequate blood pressure, targeting a systolic BP of greater than or equal to 100 mmHg in an adult, and greater than or equal to the normal predicted systolic BP in a child.
- Acquire a 12 lead ECG. Treat as per the appropriate guideline if STEMI or dysrhythmia is present.
- Utilise the post cardiac arrest checklist prior to commencing transport.

If the systolic BP is lower than the target

For an adult:

- Consider administering 500 ml of 0.9% sodium chloride IV. Repeat once if required.
- Administer metaraminol IV:
  - Administer via an infusion device:
    - Administer an initial bolus of 0.5-1 mg metaraminol IV.
    - Start the infusion at 2 mg/hour and adjust the rate as required.
  - Alternatively, administer a bolus of 0.5-1 mg metaraminol IV every 5-10 minutes as required.
- Consider administering adrenaline if the blood pressure is unresponsive to metaraminol, or the patient is hypotensive and bradycardic:
  - Administer an adrenaline infusion via an infusion device. Start at a rate of 0.5 mg/hour and adjust the rate as required, or
  - Place 1 mg of adrenaline into a 1 litre bag of 0.9% sodium chloride (1:1,000,000 solution):
    - Administer as an infusion. Start at 2 drops/second and adjust the rate as required, or
    - Administer a bolus of 0.01 mg adrenaline (10 ml) IV every 1-2 minutes as required.

For a child aged 5-14 years:

- Consider administering 10 ml/kg of 0.9% sodium chloride IV. Repeat once if required.
- Administer a bolus of metaraminol IV (see the paediatric drug dose tables) every 5-10 minutes as required.
- Consider administering adrenaline if the blood pressure is unresponsive to metaraminol or the patient is hypotensive and bradycardic:

Treatment following ROSC

- Administer an adrenaline infusion via an infusion pump using service guidelines, or
- Place 1 mg of adrenaline into a 1 litre bag of 0.9% sodium chloride (1:1,000,000 solution):
  - Administer as an infusion. Start at 1 drop/second and adjust the rate as required, or
  - Administer IV boluses (see the paediatric drug dose tables) every 1-2 minutes as required.

For a child aged less than five years:

- Consider administering 10 ml/kg of 0.9% sodium chloride IV. Repeat once if required.
- Administer a bolus of metaraminol IV (see the paediatric drug dose tables) every 5-10 minutes as required.
- Consider administering adrenaline IV if the blood pressure is unresponsive to metaraminol or the patient is hypotensive and bradycardic.
  - Administer an adrenaline infusion via an infusion pump using service guidelines, or
  - Administer IV boluses (see the paediatric drug dose tables) every 1-2 minutes as required.

Backup

- Backup from an ICP/CCP should be requested.

Referral and transport

- Transport to a hospital with PCI facilities if the patient has STEMI provided this is feasible and safe.
- Transport to a hospital with intensive care facilities if the patient is not obeying commands, provided this is feasible and safe.

Post cardiac arrest checklist

- ✓ Airway and ventilation are adequate/secure.
- ✓ Vital signs, including blood glucose level.
- ✓ Backup for RSI called for if indicated.
- ✓ IV access secure.
- ✓ $ETCO_2$ 35-45 mmHg if ventilated.
- ✓ Sedation and neuromuscular blockade if indicated.

Treatment following ROSC

- ✓ BP above target.
- ✓ 12 lead ECG.
- ✓ Confirm appropriate transport destination.

Additional information

General principles

- Return of spontaneous circulation (ROSC) is the presence of a palpable pulse, or clear signs of spontaneous circulation (such as non-agonal breathing, normal $ETCO_2$ or active movement) in the absence of CPR.
- The most important aspects of post cardiac arrest care are to maintain an adequate airway, breathing and circulation, avoiding hypoxia, hyperventilation and hypotension.
- In the first few minutes following ROSC, the patient's physiological state is often quite unstable and it is important to monitor the patient closely and intervene as required. It is appropriate to prepare for transport during this time and to complete the post cardiac arrest checklist, but do not immediately commence transport unless the patient has an unresolved time critical problem.

Targeted temperature management

- Targeted temperature management (TTM) and the avoidance of hyperthermia may improve outcomes following cardiac arrest in patients who remain unconscious.
- If core temperature monitoring is available, target a temperature of 35-36°C. Following ROSC, most patients covered with a single blanket will achieve a temperature of approximately 36°C. If core temperature monitoring is not available, cover the patient with a blanket.
- It is useful to document the patient's tympanic temperature post ROSC, but this is not a priority.
- There is no role for active warming unless the patient has had a cardiac arrest secondary to hypothermia.
- There is no role for active cooling unless the patient has hyperthermia.

Blood pressure

- There is reduced blood flow to the brain post cardiac arrest, even if blood pressure is normal.
- Any fall in blood pressure will further reduce blood flow to the brain and this is why a systolic blood pressure greater than 100 mmHg is targeted for an adult, and a blood pressure above the normal

Treatment following ROSC

predicted systolic blood pressure is targeted for a child.

Response and/or transport by helicopter

- Response by helicopter may be indicated if the patient has sustained ROSC and the helicopter will arrive on scene at least 15 minutes sooner than a road-based CCP.
- Transport by helicopter may be indicated if the patient has sustained ROSC and it will save more than 30 minutes compared with road transport.
- For helicopter transport criteria to be met, the road transport time will usually need to be more than 90 minutes.

More Information

Version history

16 June 2026 1.1.0.1
Update: Post-resuscitation treatment guidance updated.

11 September 2023 1.0.4
Update: Multiple changes.